

DASHBOARD DESIGN USING CAN BUS

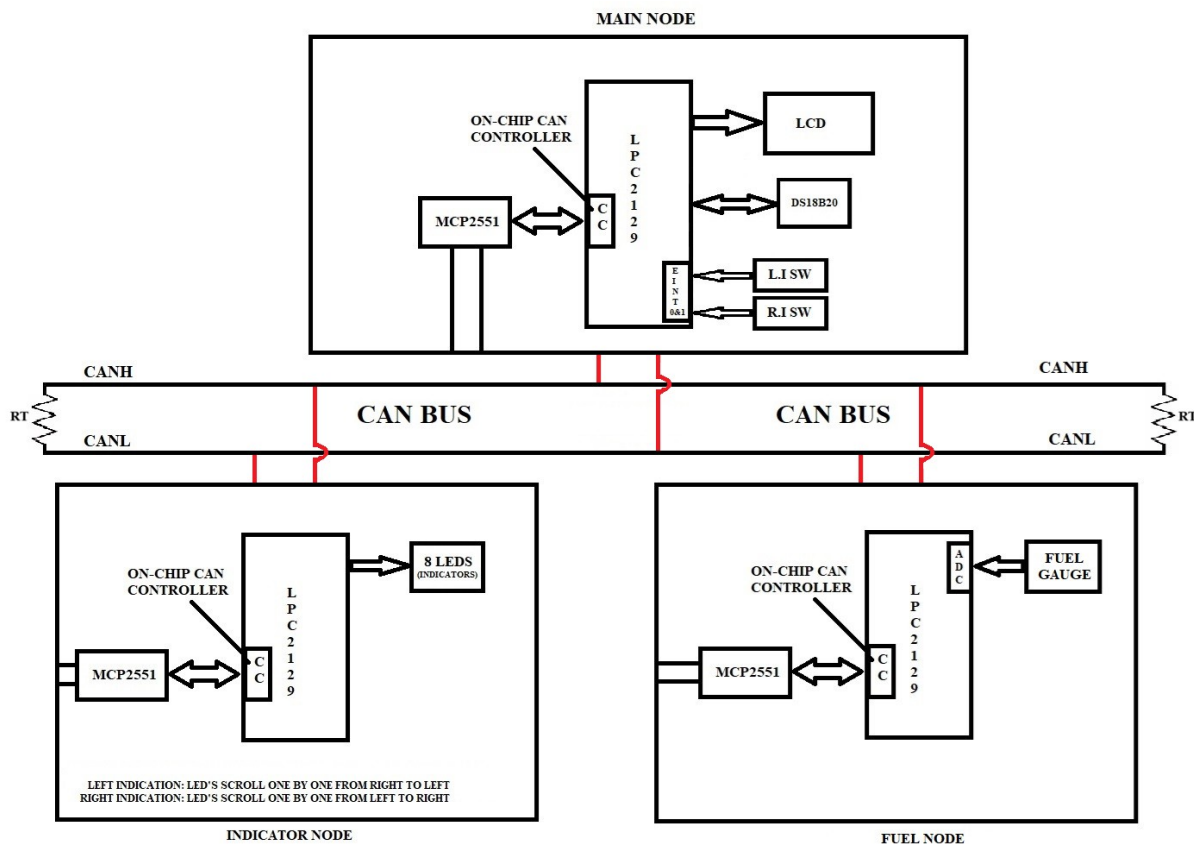
AIM:

The main aim of this project is to display the engine temperature, fuel percentage and give the left and right indicators using CAN protocol.

INSIGHT:

- Knowledge of Embedded-C programming
- Thorough with the LPC2129 Architecture, General purpose I/O Interface, ADC & CAN interface
- Understanding of CAN protocol

BLOCK DIAGRAM:



HARDWARE REQUIREMENTS:

- LPC 2129
- CAN Transceiver (MCP2551)
- LEDS
- LCD
- Switches
- DS18B20 Temperature Sensor
- Fuel Guage
- USB to UART Converter

SOFTWARE REQUIREMENTS:

- 1) EMBEDDED C – PROGRAMMING
- 2) KEIL-C COMPILER
- 3) FLASH MAGIC

SEQUENCE TO BE FOLLOWED FOR IMPLEMENTATION:

- Create New Folder in your server/laptop/PC and save that folder with your project name
- Individually can check each module.
- First check lcd to display character constant, string constant and integer constant.
- Next check the on-chip ADC by connecting variable voltage (from potentiometer) and display input value on LCD.
- Then develop logic for reading the Fuel gauge sensor information by using on-chip ADC and display it (Fuel %) on LCD.
- Then develop logic for reading the temperature sensor (Engine temperature) information and display it (engine temperature) on LCD. (refer LMS for DS18B20 temperature working code)
- Then check the external interrupt (EINT0 & EINT1) working condition with the help of basic program.
- Then refer the LMS for on-chip RTC working code and test it on hardware board.
- Then download the CAN basic code from LMS and test it on hardware boards.
- Once all the above modules are checked then start preparing the main code for each node.

- MAIN NODE: In continuous loop, Read the temperature information and display it on LCD. And based on the interrupt generated, need to send the indicator signal to the indicator node and display the indicators status also on LCD. And one more task of this main node is receiving the fuel node information and display it on LCD.
- INDICATOR NODE: Continuously waiting for the data from the main NODE using CAN. If any data is received from main NODE, based on that it is controlling the indicator signals connected to microcontroller. Refer the block diagram for how to give the indicator signals with the help of LED'S.
- FUEL NODE: Continuously reading the Fuel gauge sensor information with the help of on chip ADC and send that information to the main node.
- If you're getting this output, then your project is completed.

***** ALL THE BEST *****